

Probepriifung - Diskussion

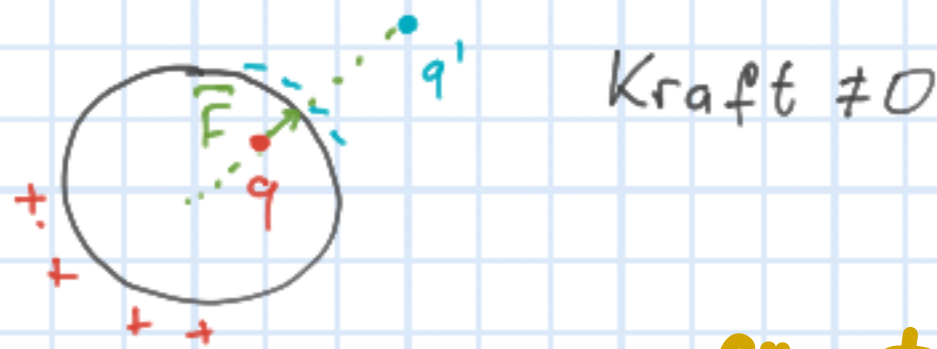
1.1) $\vec{F}(\vec{R}) \rightarrow$ zentrales Feld

$$\boxed{\text{rot } \vec{F} = \vec{0}}$$

↳ Formel für Kugel mit zentralem Feld

ℳ El. Feld einer Kugel mit Radius $r=r_0$

1.2)



Kraft $\neq 0$

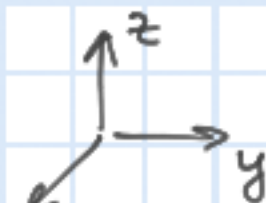
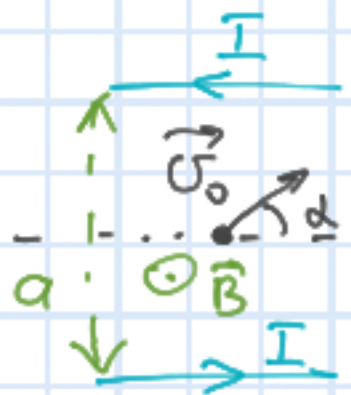
1.3)

Ja: Dirichlet. - Randproblem

$\Rightarrow \Phi \leftarrow$ eindeutig $\Rightarrow \vec{E} = -\vec{\nabla} \Phi$ auch eindeutig

↳ Potentiale auf Oberfläche wissen

1.4)



unabhängig von d

$$\vec{B} \perp \vec{U}_0$$

$$\Rightarrow |\vec{F}| \propto |\vec{U}_0 \times \vec{B}| = |\vec{U}_0| |\vec{B}|$$

(nur ganz in der Mitte) m-Feld $\neq$ konst.

1.5)



egal was in der Kugel ist

$$\text{Fluss} = \oint_S \vec{B} \cdot d\vec{s} = \int_V \text{div } \vec{B} dV = 0$$

↳ Gauß. Satz $\parallel 0$ (2. Maxw. Gl.)

Oberfläche $\rightarrow$ Volumen

1.6)



$$\vec{N} = \vec{m} \times \vec{B}$$

$$|\vec{N}_1| = |\vec{N}_2| > |\vec{N}_3|$$

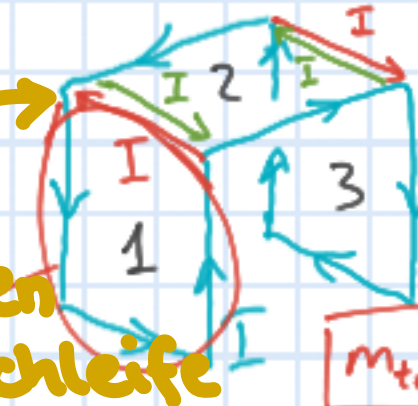
$$\vec{m} \perp \vec{I}$$

$$(\vec{m} = I \vec{A})$$

↳ kann wohl auch mit rechter Handregel begründet werden

1.7)

Trick: wir schließen flache Schleife



$$m_{\text{tot}, y} = 0$$

$$\vec{m}_{\text{tot}} = \vec{m}_1 + \vec{m}_2 + \vec{m}_3$$

$$\vec{m}_2 \perp \hat{y} \Rightarrow m_{2y} = 0$$

$$\vec{m}_1 = -\vec{m}_2 \Rightarrow (\vec{m}_1 + \vec{m}_2)_y = 0$$

$$\vec{m} = \frac{1}{2} \int \vec{r} \times \vec{J}(\vec{r}) d^3r$$

↳ Trick: nur flache Schleife

$$= I \int \frac{1}{2} \vec{r} \times d\vec{e} = I \vec{S}$$

1.8) $\vec{A}'(\vec{r}) = \vec{A}(\vec{r}) + \underbrace{a(\vec{r} \cdot \hat{x}) \hat{x}}_{\vec{\nabla} \chi(\vec{r})}$

$$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \hat{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$a(\vec{r} \cdot \hat{x}) \hat{x} = ax \hat{x} = \begin{pmatrix} ax \\ 0 \\ 0 \end{pmatrix} = \vec{\nabla} \left(\frac{ax^2}{2} \right)$$

Eichttransformation $\Rightarrow$ Antwort = Ja

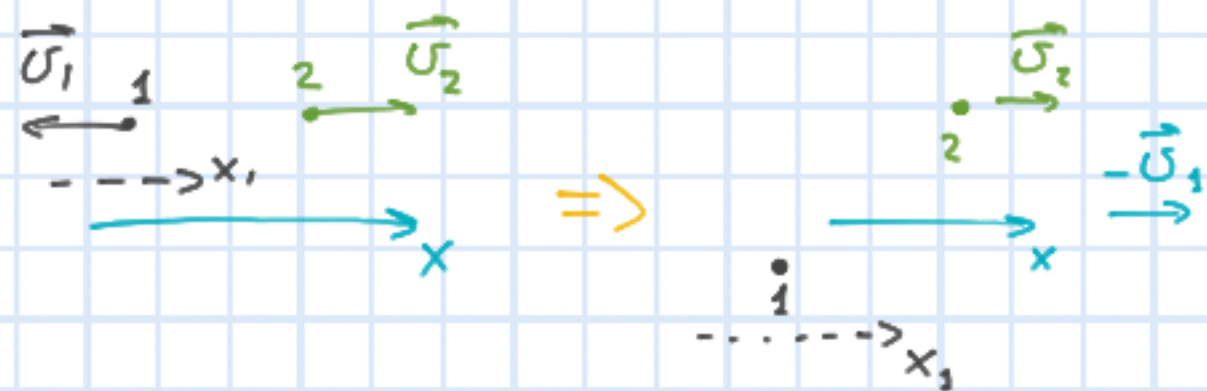
1.9) $\frac{1}{c^2} \frac{\partial^2 f}{\partial t^2} - \vec{\nabla}^2 f = u(t, \vec{r})$

Greensche Funktion in Klausuren wohl immer easy...

Antwort = Nein $\Leftarrow \frac{1}{c^2} \frac{\partial^2 G(t, \vec{r}, t', \vec{r}')}{\partial t^2} - \vec{\nabla}_r^2 G(t, \vec{r}, t', \vec{r}') = -4\pi \delta(\vec{r} - \vec{r}') \delta(t - t')$

$$f(t, \vec{r}) = -\frac{1}{4\pi} \int dt' \int d^3r' G(t, \vec{r}, t', \vec{r}') u(t', \vec{r}')$$

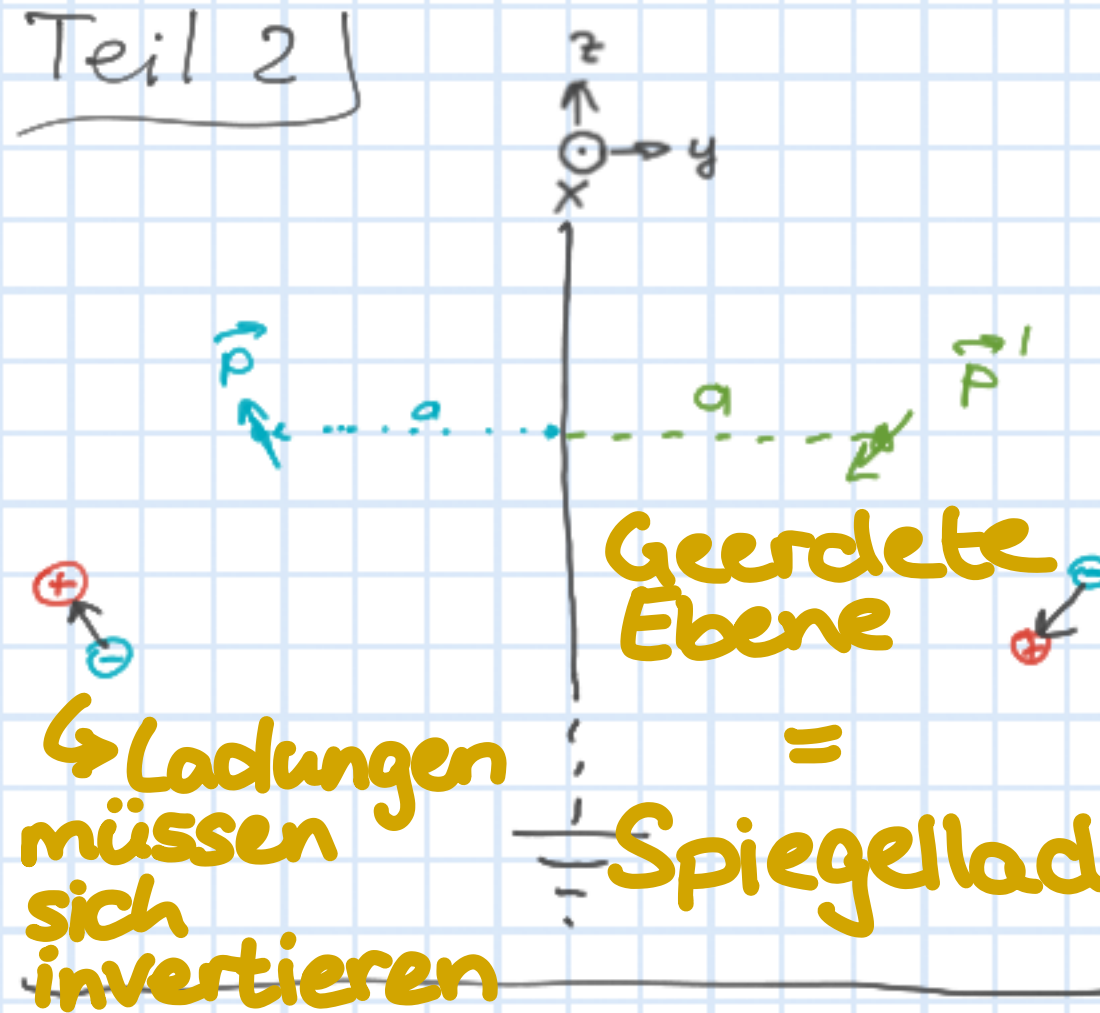
1.10



$$\left[(u_2) \right]_1 = \frac{|u_1| + u_2}{1 + \frac{|u_1| u_2}{c^2}} = \frac{\frac{c}{2} + \frac{c}{2}}{1 + \frac{c^2/4}{c^2}} = c \frac{1}{1 + \frac{1}{4}} = \frac{4}{5} c$$

$$|u_1| = |u_2| = \frac{c}{2}$$

Teil 2



Projektion Pfeil auf die Achsen

$$\vec{p} = \begin{pmatrix} 0 \\ p_y \\ p_z \end{pmatrix}$$

$$\vec{r} = (x, y, z)$$

$$\vec{p}' = \begin{pmatrix} 0 \\ p_y \\ -p_z \end{pmatrix}$$

$$\vec{r}' = (x', y', z') = (x, -y, z)$$

$$\Rightarrow \vec{r} - \vec{r}' = \begin{pmatrix} 0 \\ 2y \\ 0 \end{pmatrix} \quad (*)$$

$$W_{\vec{p}, \vec{p}'} = -\vec{E}_{\text{dip}, \vec{p}} \cdot \vec{p}' = \frac{1}{4\pi\epsilon_0} \frac{(\vec{p} \cdot \vec{p}') |\vec{r} - \vec{r}'|^2 - 3 [(\vec{r} - \vec{r}') \cdot \vec{p}] [(\vec{r} - \vec{r}') \cdot \vec{p}']}{|\vec{r} - \vec{r}'|^5}$$

$$\vec{F}_{\vec{p}} = -\vec{\nabla}_{\vec{r}} W_{\vec{p}, \vec{p}'} = -\frac{1}{4\pi\epsilon_0} \vec{\nabla}_{\vec{r}} \frac{f(\vec{r})}{g(\vec{r})} = -\frac{1}{4\pi\epsilon_0} \left(\frac{1}{g(\vec{r})} \vec{\nabla}_{\vec{r}} f(\vec{r}) - \frac{f(\vec{r})}{g^2(\vec{r})} \vec{\nabla}_{\vec{r}} g \right) \quad (**)$$

$$f(\vec{r}) = (\vec{p} \cdot \vec{p}') [(x-x')^2 + (y-y')^2 + (z-z')^2] - 3 [(\vec{r} - \vec{r}') \cdot \vec{p}] [(\vec{r} - \vec{r}') \cdot \vec{p}']$$

$$\vec{\nabla} f(\vec{r}) = (\vec{p} \cdot \vec{p}') 2(\vec{r} - \vec{r}') - 3 \vec{p} [(\vec{r} - \vec{r}') \cdot \vec{p}'] - 3 [(\vec{r} - \vec{r}') \cdot \vec{p}] \vec{p}' \quad (***)$$

$$g(\vec{r}) = [(x-x')^2 + (y-y')^2 + (z-z')^2]^{\frac{5}{2}} \Rightarrow \vec{\nabla} g = \frac{5}{2} |\vec{r} - \vec{r}'|^3 \cdot 2(\vec{r} - \vec{r}') \quad (****)$$

(***) & (****) $\rightarrow$ (**):

$$\vec{F}_{\vec{p}} = -\frac{1}{4\pi\epsilon_0} \left\{ \frac{(\vec{p} \cdot \vec{p}') \cdot 2(\vec{r} - \vec{r}') - 3\vec{p} [(\vec{r} - \vec{r}') \cdot \vec{p}'] - 3\vec{p}' [(\vec{r} - \vec{r}') \cdot \vec{p}]}{|\vec{r} - \vec{r}'|^5} \right.$$

$$\left. - \frac{(\vec{p} \cdot \vec{p}') (\vec{r} - \vec{r}')^2 - 3 [(\vec{r} - \vec{r}') \cdot \vec{p}] [(\vec{r} - \vec{r}') \cdot \vec{p}']}{|\vec{r} - \vec{r}'|^7} \right\} \quad 5 |\vec{r} - \vec{r}'|^3 (\vec{r} - \vec{r}')$$

(*): $\vec{r} - \vec{r}' = \begin{pmatrix} 0 \\ 2y \\ 0 \end{pmatrix}$

$$\vec{p} \cdot \vec{p}' = \begin{pmatrix} 0 \\ p_y \\ p_z \end{pmatrix} \cdot \begin{pmatrix} 0 \\ p_y \\ -p_z \end{pmatrix} = p_y^2 - p_z^2$$

$$F_{\vec{p}} = -\frac{1}{4\pi\epsilon_0} \left\{ \frac{2(p_y^2 - p_z^2) \begin{pmatrix} 0 \\ 2y \\ 0 \end{pmatrix} - 3 \cdot \cancel{2y} p_y \begin{pmatrix} 0 \\ p_y \\ p_z \end{pmatrix} - 3 \cdot \cancel{2y} p_y \begin{pmatrix} 0 \\ p_y \\ -p_z \end{pmatrix}}{|2y|^7} \cdot (-1) \right.$$

$$\left. - \frac{(p_y^2 - p_z^2) \cancel{4y^2} - 3 \cdot \cancel{2y} p_y \cdot \cancel{2y} p_y}{|2y|^7} \cdot \begin{pmatrix} 0 \\ 2y \\ 0 \end{pmatrix} (-1) \right\}$$

$$y = -a < 0$$

$\Downarrow$

Kürzung

$$\frac{y}{|y|^5} = -\frac{1}{y^4}$$

$$\frac{y^3}{|y|^7} = -\frac{1}{y^4}$$

Nur $F_y \neq 0$

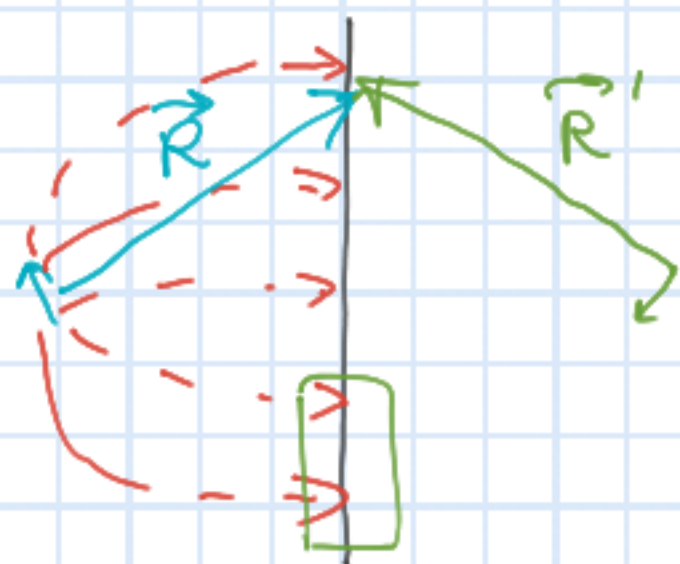
(a)

$$F_y = \frac{1}{4\pi\epsilon_0} \left\{ \frac{2(p_y^2 - p_z^2) - 3p_y^2 - 3p_z^2}{(2y)^4} - \frac{(p_y^2 - p_z^2 - 3p_y^2) \cdot 5}{(2y)^4} \right\} =$$

$$= \frac{-3(p_y^2 - p_z^2) - 6p_y^2 + 15p_z^2}{64\pi\epsilon_0 y^4} = \frac{6p_y^2 + 3p_z^2}{64\pi\epsilon_0 a^4}$$

$y = -a$

(b)



$\rightarrow y$

$$\frac{\rho}{\epsilon_0} = -E_y$$

$$\vec{E} = \vec{E}_p + \vec{E}_{p'} = \frac{1}{4\pi\epsilon_0} \left[\frac{3\vec{R}(\vec{p} \cdot \vec{R}) - \vec{p}(\vec{R}^2)}{|\vec{R}|^5} + \frac{3\vec{R}'(\vec{p}' \cdot \vec{R}') - \vec{p}'(\vec{R}'^2)}{|\vec{R}'|^5} \right]$$

$$E_x = E_z = 0$$

$$E_y = \frac{1}{4\pi\epsilon_0} \left[\frac{3R_y(\vec{p} \cdot \vec{R}) - p_y \vec{R}^2}{|\vec{R}|^5} + \frac{3R'_y(\vec{p}' \cdot \vec{R}') - p'_y \vec{R}'^2}{|\vec{R}'|^5} \right] \quad \text{nächste Seite} \rightarrow$$

$$|\vec{R}|^3 = |\vec{R}'|^3 = R^3$$

$$p'_y = p_y$$

$$R'_y = -R_y$$

 $\Rightarrow$

$$(\vec{p} \cdot \vec{R}) = p_y R_y + p_z R_z$$

$$(\vec{p}' \cdot \vec{R}') = p_y (-R_y) + (-p_z) R_z = -(\vec{p} \cdot \vec{R})$$

$$\rightarrow E_y = \frac{1}{4\pi\epsilon_0} \left[\frac{6R_y(\vec{p} \cdot \vec{R}) - 2p_y \vec{R}^2}{R^5} \right] = \frac{1}{4\pi\epsilon_0} \frac{6a(p_x x + p_y a + p_z z) - 2p_y(x^2 + a^2 + z^2)}{(x^2 + a^2 + z^2)^{5/2}}$$

$$R_y = a$$

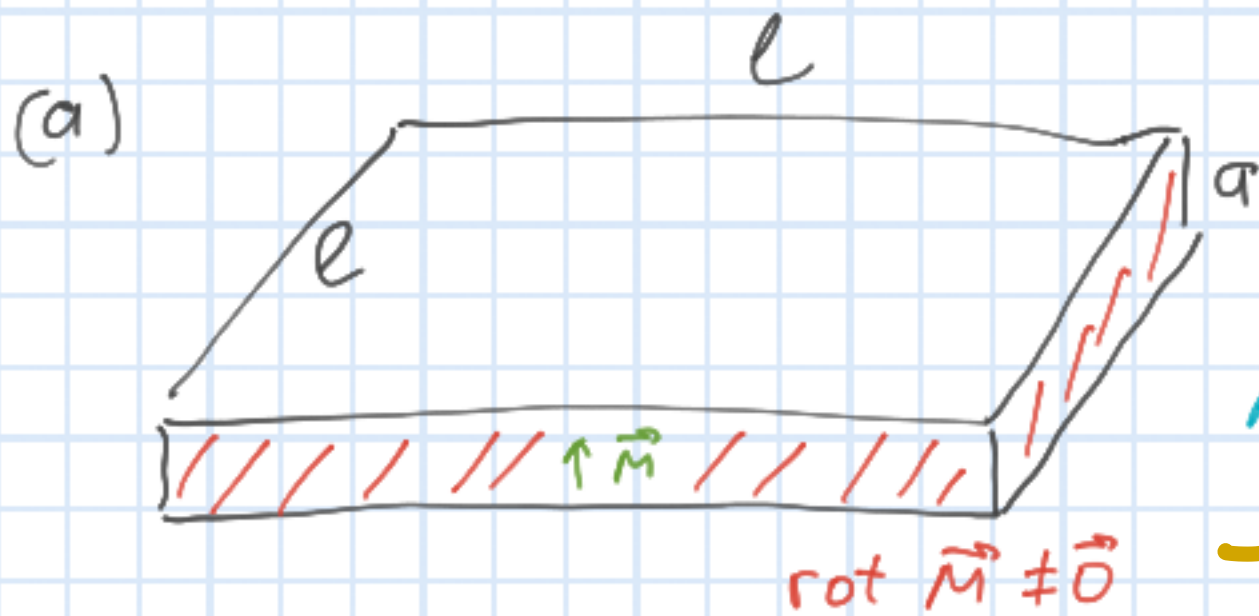
$$\vec{R} = \begin{pmatrix} x \\ a \\ z \end{pmatrix} : \quad R^2 = x^2 + a^2 + z^2$$

$$(\vec{p} \cdot \vec{R}) = p_x x + p_y a + p_z z$$

$$\frac{\sigma}{\epsilon_0} = -E_y \Rightarrow \sigma(x, z) = -\frac{1}{4\pi} \frac{6a(p_x x + p_z z + p_y a) - 2p_y(x^2 + z^2 + a^2)}{(x^2 + z^2 + a^2)^{5/2}}$$

Teil 3

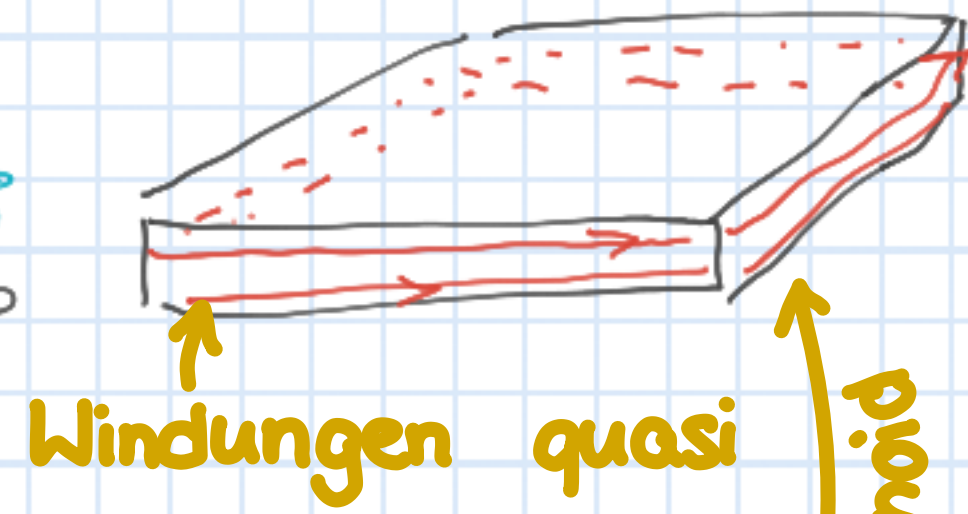
$$l \gg a$$



$\text{rot } \vec{M} = 0$
 $\text{rot } \vec{M} \neq \vec{0}$

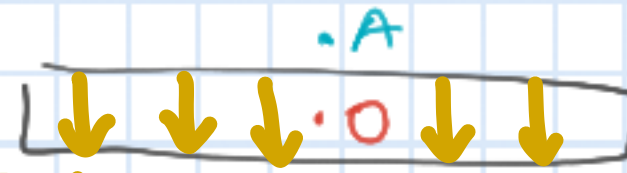
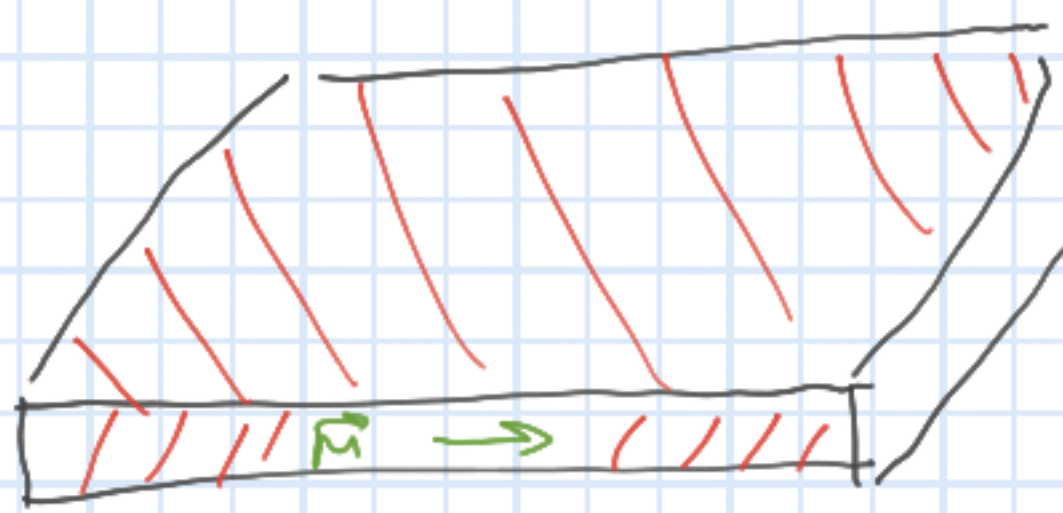
Antwort: $\vec{B}_0, \vec{B}_A \propto \frac{1}{l} \rightarrow \vec{0}$
 $\frac{a}{l} \rightarrow 0$

→ Biot-Savart Gesetz



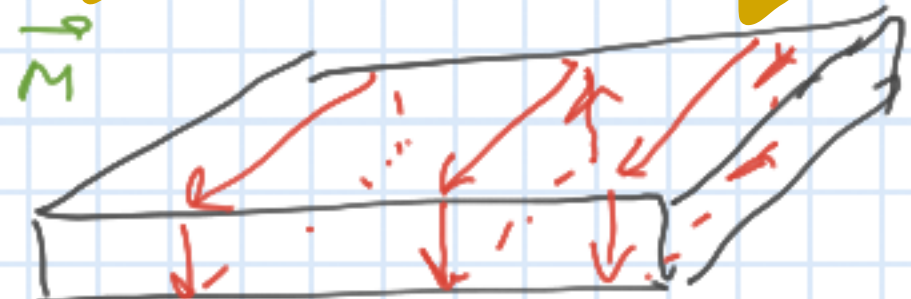
≡ Solenoid

(b)



Antwort: $\vec{B}_0 \rightarrow \mu_0 \vec{M}$
 $\vec{B}_A = \vec{0}$

$IN = M$



keine freien Ströme $\Rightarrow \vec{\nabla} \times \vec{B} = \mu_0 \vec{\nabla} \times \vec{M}$

$\Downarrow$
wie Solenoid $N I dz = M dz \leftarrow \text{HA 10}$

unendl. lang. Sol.

$\vec{B} = \mu_0 I N = \mu_0 \vec{M}$

Formeln müssen in Klausur nicht neu hergeleitet werden

Lösungen HA 10

Problem 1: Jackson 5.19

(a) keine Ströme der freien Ladungen

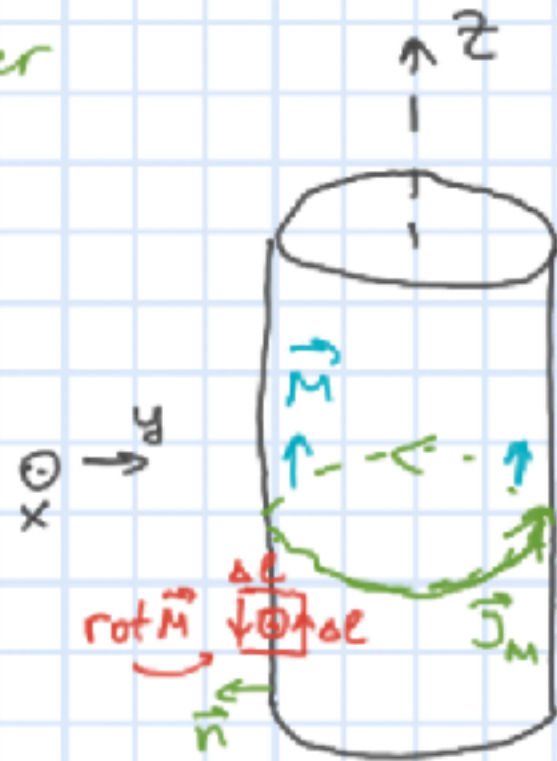
$$\vec{J} = \vec{0}$$

$\Downarrow$

$$\text{rot } \vec{H} = \vec{0}$$

$$\vec{\nabla} \times \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right) = \vec{0}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{\nabla} \times \vec{M} = \mu_0 \vec{J}_M$$



$\vec{J}_M \rightarrow 2D$ Flächenstromdichte $\sim$ Solenoid (Pr. 1, HA 9) (Jackson 5.3)

Gesamtstrom durch dz :

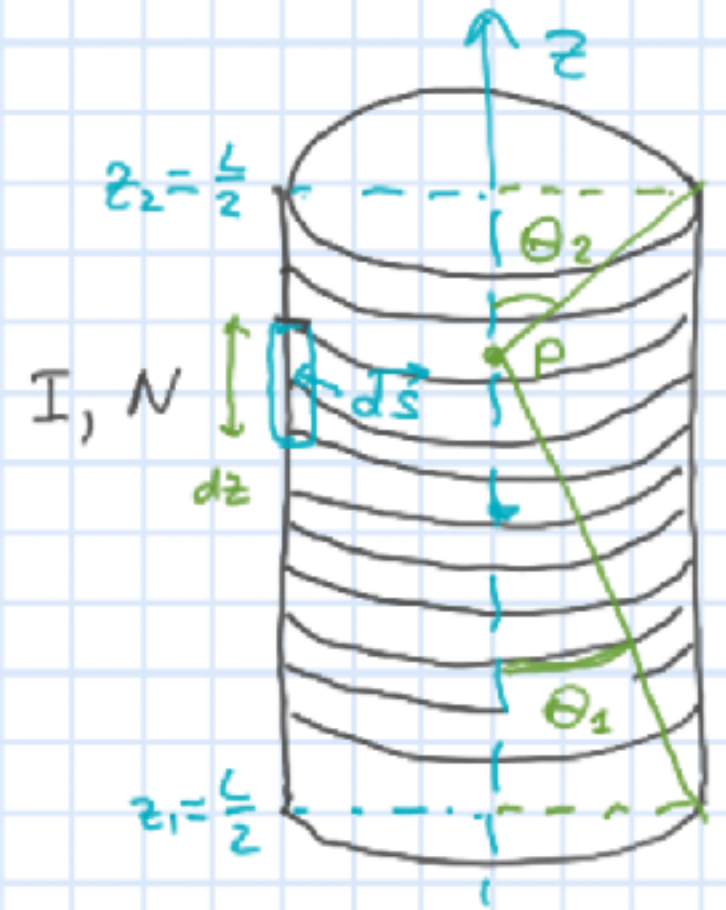
$$I N dz = \int \vec{J}_M \cdot d\vec{s} = |\vec{M} \times \vec{n}| dz$$

$$I N = M$$

DIE wichtige Formel hier

HA 9, P.1:

$$B_z = \frac{\mu_0 I N}{2} (\cos \theta_1 + \cos \theta_2) \quad (*)$$



$$\vec{\nabla} \times \vec{M} = \begin{cases} \vec{0} & \text{außerhalb} \\ \vec{0} & \text{innerhalb} \\ ? & \text{Grenze} \end{cases}$$

Stockes. S: $\text{rot } \vec{M} \cdot \Delta l^2 \hat{x} = M \Delta l$

$$\text{rot } \vec{M} = \frac{M \hat{x}}{\Delta l} = \frac{\vec{M} \times \vec{n}}{\Delta l} \xrightarrow{\Delta l \rightarrow 0} \infty \rightarrow \delta(y-y_0) \cdot \text{konst}$$

$$\int_{-\Delta l/2}^{\Delta l/2} \text{rot } \vec{M} dy = \frac{\vec{M} \times \vec{n}}{\Delta l} \Delta l \Big|_{\Delta l \rightarrow 0} = \vec{M} \times \vec{n}$$

Teil 4

$$\vec{p}(t) = \operatorname{Re} [\vec{p}_\omega e^{-i\omega t}]$$

$$\vec{n}_r = \frac{\vec{r}}{r} = \vec{n}$$

$$\vec{H}_\omega = \frac{1}{4\pi} c k^2 \underbrace{\frac{e^{ikr}}{r} \left(1 - \frac{1}{ikr}\right)}_{\vec{F}(\vec{r})} \vec{n}_r \times \vec{p}_\omega$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

3. Maxw. Gl. : $\vec{E}_\omega = \frac{i}{k} \sqrt{\frac{\mu_0}{\epsilon_0}} \vec{\nabla} \times \vec{H}_\omega = i \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{1}{4\pi} \frac{1}{\sqrt{\mu_0 \epsilon_0}} k \vec{\nabla} \times (\vec{F}(\vec{r}) \times \vec{p}_\omega) = \vec{\nabla}$

aus der Aufgabe

$$\vec{\nabla} \times (\vec{F} \times \vec{p}) = \epsilon_{ijk} \nabla_j \epsilon_{k\ell m} \overset{\text{HAI}}{F_\ell} p_m = (\delta_{ie} \delta_{jm} - \delta_{im} \delta_{je}) \nabla_j F_\ell p_m =$$

$$= \nabla_j F_i p_j - \nabla_j F_j p_i = (\vec{p} \cdot \vec{\nabla}) \vec{F} - \vec{p} \cdot (\vec{\nabla} \cdot \vec{F}) \quad (1)$$

Nabla bleibt IMMER
links von F

$$\vec{F}(\vec{r}) = f(r) \cdot \vec{r} = \underbrace{\frac{e^{ikr}}{r^2} \left(1 - \frac{1}{ikr}\right)}_{f(r)} \vec{r} \quad (2)$$

das ist f (für Formel (5) wichtig)

Nebenrechnung

Kettenregel

$$\nabla_j F_i = \nabla_j f(r) r_i = \frac{df}{dr} \underbrace{\frac{\partial r}{\partial r_j}}_{\substack{\text{keine} \\ \text{Summe}}} r_i + f(r) \delta_{ij} = \frac{1}{r} \frac{df}{dr} r_j r_i + f(r) \delta_{ij} \quad (3)$$

divergenz
berechnen

$$\nabla_j F_j = r \frac{df}{dr} + f(r) \cdot 3 \quad (4)$$

Summe!

$$\begin{aligned} \frac{df}{dr} &= \frac{e^{ikr}}{r^2} \left(ik - \frac{2}{r} \right) \left(1 - \frac{1}{ikr} \right) + \frac{e^{ikr}}{r^2} (-1)(-1) \frac{1}{ikr^2} = \frac{e^{ikr}}{r^2} \left(ik - \frac{2}{r} - \frac{1}{r} + \frac{2}{ikr^2} + \frac{1}{ikr^2} \right) = \\ &= \frac{e^{ikr}}{r^2} \left(ik - \frac{3}{r} + \frac{3}{ikr^2} \right) \quad (5) \end{aligned}$$

$$\begin{aligned} \vec{p}(\vec{\nabla} \cdot \vec{F}) &\stackrel{(4)(5)}{=} \vec{p} \left[\frac{e^{ikr}}{r} \left(ik - \frac{3}{r} + \frac{3}{ikr^2} \right) + 3 \frac{e^{ikr}}{r^2} \left(1 - \frac{1}{ikr} \right) \right] = \\ &= \vec{p} \frac{e^{ikr}}{r} \left[ik - \cancel{\frac{3}{r}} + \frac{3}{ikr^2} + \cancel{\frac{3}{r}} - \cancel{\frac{3}{ikr^2}} \right] = \vec{p} \frac{e^{ikr}}{r} ik \end{aligned}$$

Das ist ein Term
in dieser
Geschichte

hier der andere

$$(\vec{p} \cdot \vec{\nabla}) \vec{F} = (\vec{p} \cdot \vec{r}) \cdot \vec{r} \cdot \frac{e^{ikr}}{r^3} \left(ik - \frac{3}{r} + \frac{3}{ikr^2} \right) + \frac{e^{ikr}}{r^2} \left(1 - \frac{1}{ikr} \right) \vec{p} =$$

$$= \frac{e^{ikr}}{r} \frac{\vec{n}_r = \frac{\vec{r}}{r}}{ik} (\vec{p} \cdot \vec{n}_r) \vec{n}_r - \frac{e^{ikr}}{r^2} \left(1 - \frac{1}{ikr} \right) (\vec{p} - 3(\vec{p} \cdot \vec{n}_r) \vec{n}_r)$$

↑
(3), (4), (5) statt $\vec{r} \rightarrow \vec{n}_r$

$$(1) \Rightarrow (\vec{p} \cdot \vec{\nabla}) \vec{F} - \vec{p} (\vec{\nabla} \cdot \vec{F}) = \frac{e^{ikr}}{r} ik [(\vec{p} \cdot \vec{n}_r) \vec{n}_r - \vec{p}] + \frac{e^{ikr}}{r^2} \left(1 - \frac{1}{ikr} \right) (\vec{p} - 3(\vec{p} \cdot \vec{n}_r) \vec{n}_r)$$

Faradaysches Gesetz

$$\vec{E}_\omega = \frac{1}{4\pi\epsilon_0} \left[\frac{e^{ikr}}{r} (-k^2) [(\vec{p} \cdot \vec{n}_r) \vec{n}_r - \vec{p}] + \frac{e^{ikr}}{r^2} \left(ik - \frac{1}{r} \right) [\vec{p} - 3(\vec{p} \cdot \vec{n}_r) \vec{n}_r] \right]$$

Strahlung / Relativitätstheorie

$$\vec{E}_\omega = \frac{1}{4\pi\epsilon_0} \left\{ \frac{e^{ikr}}{r} k^2 [(\vec{n}_r \times \vec{p}_\omega) \times \vec{n}_r] + \frac{e^{ikr}}{r^2} \left(ik - \frac{1}{r} \right) [\vec{p}_\omega - 3(\vec{p}_\omega \cdot \vec{n}_r) \vec{n}_r] \right\}$$

$$\vec{n} \times (\vec{n} \times \vec{p}) = \vec{n} (\vec{n} \cdot \vec{p}) - \vec{p} \cdot \vec{n}^2$$